Sahaj Bhadja

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Summary

Backend software engineer with 2+ years building production Python services, ETL pipelines, data models, and AI-powered workflows across AWS and Azure. Experienced in complex integrations, SQL analysis, containers, automation, and operational reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant data services on Azure SQL and Blob Storage for 500+ users, implementing row-level security, normalized models, validation, and indexed queries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant AI orchestration for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and production monitoring, improving throughput by 3x while preventing cascading pipeline failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered production Python services and REST integrations handling 10,000+ daily requests, combining secure ingestion, AWS RDS persistence, and multimodal AI workflows while maintaining stable performance during peak usage.
- Implemented concurrent ETL-style processing for 1,000+ files with validation, error isolation, structured logging, automated tests, and operational reporting, reducing processing turnaround by 60% for 150+ users.

USC AD Labs **Los Angeles, CA**
Machine Learning Researcher *January 2025 – May 2025*
- Built Python data and simulation pipelines processing 100,000+ trajectory logs, improving long-horizon autonomous-driving planning by 18% through offline reinforcement learning, reproducible evaluation, and structured experiment tracking.
- Architected a spatial-temporal encoder combining HD-map geometry and agent trajectories through multi-head attention, improving closed-loop stability by 15% while generating explainable visual analysis of production model behavior.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform microservices with Jenkins pipelines, automated VM builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening integration quality and operational production reliability.
- Optimized ETL and API workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving response times by 30% and data-pipeline efficiency by 68% for downstream consumers.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for cross-platform booking workflows spanning reservations, scheduling, status updates, and payments across 50,000+ records, integrating Firebase persistence with real-time event synchronization and validation.
- Implemented webhook-driven payment processing, Firebase transactions, event synchronization, and analytical micro-batches, reducing manual payment handling by 70% while preventing duplicate bookings and improving confirmation speed by 40%.

Skills

Languages & Data: Python, SQL, Java, JavaScript, TypeScript, Bash, complex data structures, data modeling
ETL & Backend: ETL pipelines, data integration, FastAPI, Flask, REST APIs, microservices, background workers, webhooks
Cloud & Infrastructure: AWS, Azure, GCP, Amazon S3, AWS Glue, AWS RDS, Docker, Kubernetes, Linux
Databases & Analytics: PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB, Firebase, Jupyter, query optimization
AI & Operations: LLM systems, agentic workflows, RAG, PyTorch, observability, metrics, logs, root-cause analysis, automation
Engineering: Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, Agile, documentation, customer requirements

Projects

Big-Brain | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an AI-powered retrieval platform over 50,000+ documents with contextual search, summarization, secure access controls, and sub-second similarity queries, reducing document review time by 60% through grounded workflows.

Linguistix | *Python, AWS, Docker, CodePipeline, REST APIs*
- Developed a containerized data-processing service with automated releases, validation checks, structured error handling, and scalable REST interfaces, achieving 87% accuracy while strengthening reproducible deployment and production reliability.

FinNexus | *TypeScript, Kubernetes, MongoDB, WebSockets, Plaid*
- Built secure real-time integrations supporting 50,000+ users and 1M+ transactions, combining token-based APIs, persistent filtering, WebSocket synchronization, and Kubernetes-backed deployment automation across high-volume financial data workflows.